

Worked Solutions

Digital Systems and Computer Organization

Set B: Original | CSE Semester 3

Companion to dsco-original.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. K-map cell count

Correct option: (C). There is one cell per input combination, so $2^4 = 16$ cells.

2. K-map adjacency

Correct option: (A). Gray-code ordering makes adjacent cells differ in exactly one variable.

3. SOP grouping

Correct option: (D). Groups contain a power of two cells; 8 is the only power of two listed.

4. Function reduction

Correct option: (B). Every listed minterm has $B = D$. Thus $F = \neg B \neg D + BD$, independent of A and C.

5. NAND inverter

Correct option: (C). Connecting both inputs to X gives $\neg(X \cdot X) = \neg X$.

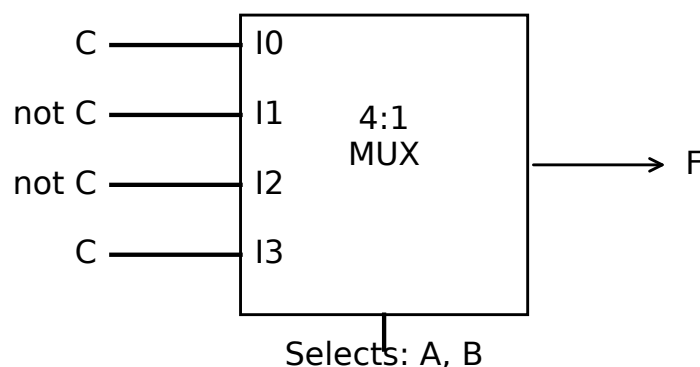
6. Signed results and overflow

Operation	Binary addition	Result	Interpretation	Overflow
11+7	01011 + 00111	10010	-14	Yes
-9+5	10111 + 00101	11100	-4	No
6-13	00110 + 10011	11001	-7	No

Only the true result 18 is outside $[-16, 15]$. For subtraction, 13 is 01101 and its two's-complement negative is 10011. A carry out by itself is not the signed-overflow flag.

7. MUX realization

With A and B as select bits (A most significant), $AB = 00, 01, 10, 11$ produces $C, \bar{C}, \bar{C}, C$, respectively. Thus $I_0 = C, I_1 = \bar{C}, I_2 = \bar{C}, I_3 = C$. One inverter supplies both complemented-C inputs.



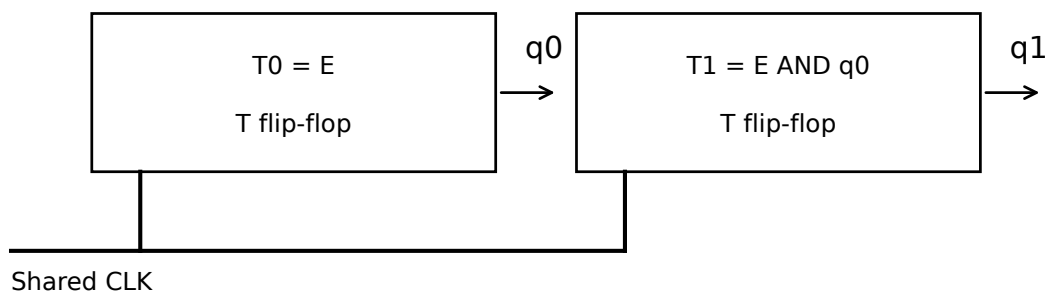
8. Enabled two-bit synchronous counter

Present q_1q_0	Next when E=0	Next when E=1
00	00	01
01	01	10
10	10	11
11	11	00

Use the T excitation equation $T_i = q_i \oplus q_i^+$. The low bit toggles exactly when E=1; the high bit toggles exactly when E=1 and the current low bit is 1.

$$T_0 = E, \quad T_1 = E q_0.$$

Feed E directly to T_0 and an AND of E and q_0 to T_1 . The clock is common.



9. Four-bit ring counter

```

module ring4(input wire clk, input wire reset,
             output reg [3:0] q);
    always @(posedge clk) begin
        if (reset) q <= 4'b0001;
        else q <= {q[2:0], q[3]};
    end
endmodule

```

On a positive clock edge with reset high, q becomes 0001. With reset low, successive states are 0010, 0100, 1000 and 0001. Concatenation moves bits 2:0 upward and puts old bit 3 into bit 0. Nonblocking assignments model sequential storage.

10. Ripple delay and decoded glitches

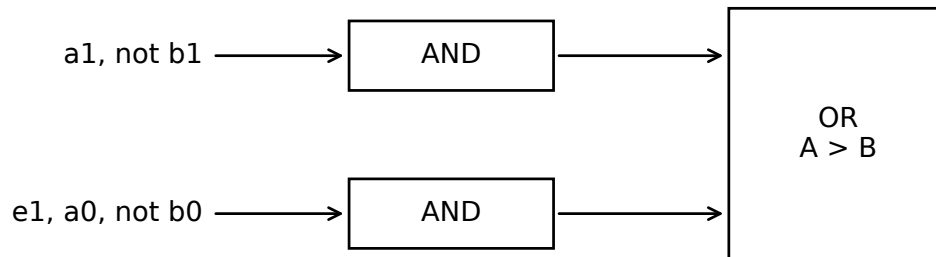
A ripple counter clocks stage 0 from the external clock and later stages from preceding flip-flop outputs. Propagation delays accumulate, so several output bits do not change simultaneously. During a transition such as 0111 to 1000, a decoder may briefly recognize an intermediate pattern. A synchronous counter drives all flip-flops from one shared clock and computes excitation logic from the old state. It avoids stage-by-stage clock ripple, although real output skew and combinational delays still require proper timing design.

11. Two-bit magnitude comparator

Let $e_i = a_i \text{ XNOR } b_i$. Then

$$G = a_1 \bar{b}_1 + e_1 a_0 \bar{b}_0, \quad E_q = e_1 e_0, \quad L = \bar{a}_1 b_1 + e_1 \bar{a}_0 b_0.$$

The upper bit decides unless equal; only then does the lower bit decide. Use XNOR gates for e_1, e_0 , inverters for complements, AND gates for product terms and OR gates to combine them. The figure gives the G network. Swap a and b for L, and AND the two XNOR outputs for E_q .



For A=10 and B=01, $e_1 = 0$, $a_1 \bar{b}_1 = 1$, so $G = 1, E_q = 0, L = 0$.

12. Memory capacity and addresses

4 KiB = 4096 bytes. Each 32-bit word occupies 4 bytes:

$$\text{words} = 4096/4 = 1024, \quad \text{byte-select bits} = \log_2(4096) = 12.$$

The word at 0x1000 occupies 0x1000, 0x1001, 0x1002 and 0x1003. The next word starts at 0x1004. The 12 bits select an offset within the 4 KiB block; the full system may use more address bits to place that block at base 0x1000. Endianness changes byte significance, not these occupied addresses.

13. Memory read and write

For a read: place the target address in MAR, assert the read control signal, wait until the memory interface indicates valid data, and latch the returned value into MDR. Transfer MDR to the destination register as required.

For a write: place the target address in MAR and outgoing data in MDR, assert write with the required timing, and wait for completion. Read moves data from memory to MDR; write moves data from MDR to memory. MAR selects the location in both cases.